

WICHITA FALLS KICKAPOO, TX, USA

WMO: 720404

Lat: 33.858N

Lon: 98.490W

Elev: 998

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00134

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	24.0	2.8	6.6	27.4	8.3	8.7	32.8	26.4	46.3	24.3	44.3	8.0	0	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	104.2	70.6	101.3	71.8	99.1	71.3	77.3	91.2	76.2	90.3	75.2	89.3	10.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	128.9	81.7	72.7	125.6	81.3	71.8	122.1	80.9	41.3	90.5	40.2	90.3	39.2	88.9	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.7	21.4	19.3	13.2	106.6	3.4	2.9	10.7	108.7	8.7	110.4	6.7	112.0	4.2	114.1	
			DB												
			WB	11.8	78.7	3.9	2.1	9.0	80.2	6.7	81.4	4.5	82.5	1.7	84.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	44.0	48.4	57.3	65.0	73.4	82.9	85.8	85.5	78.0	65.5	54.8	45.0
	DBStd	17.12	9.54	11.41	10.15	8.40	7.54	5.00	4.76	5.30	6.66	8.84	9.04	9.63
	HDD50	709	232	151	46	6	0	0	0	0	0	8	54	212
	HDD65	2565	653	473	273	100	21	0	0	0	2	103	319	621
	CDD50	6387	45	105	273	457	724	987	1109	1102	841	488	198	58
	CDD65	2768	1	7	35	101	280	537	644	637	393	118	13	2
	CDH74	34275	27	110	428	1106	2968	6785	8808	8463	4232	1170	151	27
	CDH80	18411	2	26	113	382	1318	3750	5235	5032	2117	404	29	3
(14) Wind	WSAvg	9.2	8.5	9.3	10.5	11.3	10.7	10.3	8.4	8.0	8.3	8.8	8.7	8.1
(15) Precipitation	PrecAvg	30.0	1.5	1.7	2.3	3.0	3.8	3.5	2.5	2.5	2.6	3.0	2.0	1.7
	PrecMax	48.7	3.1	4.7	5.0	8.4	16.5	9.3	7.3	8.2	9.1	6.7	7.8	6.9
	PrecMin	15.6	0.0	0.2	0.4	0.4	0.3	0.2	0.0	0.0	0.2	0.0	0.0	0.0
	PrecStd	7.2	0.9	1.4	1.4	2.1	3.1	2.0	1.9	1.9	2.0	1.8	2.0	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.9	83.1	88.4	94.8	99.6	104.5	107.7	107.9	100.5	92.9	83.6	78.6
		MCWB	54.7	57.9	61.3	65.4	69.6	70.9	70.3	69.8	70.3	68.8	64.1	58.5
	2%	DB	70.3	76.9	82.5	88.3	94.8	100.1	103.9	104.4	97.7	88.2	78.3	70.3
		MCWB	52.8	58.0	61.6	64.2	70.5	70.9	70.7	70.6	70.8	67.2	60.1	55.2
	5%	DB	65.6	72.0	79.0	84.1	90.7	97.7	100.4	102.0	94.8	84.4	73.4	65.6
		MCWB	50.4	57.4	60.6	63.4	69.3	71.6	71.9	71.8	71.2	65.9	58.9	54.3
	10%	DB	60.8	66.5	74.5	80.8	87.7	95.2	98.7	99.1	91.2	81.1	69.9	60.8
		MCWB	48.1	53.9	58.4	62.8	68.9	72.7	72.0	71.7	70.5	64.2	58.4	50.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.3	65.2	68.4	71.3	76.5	78.9	78.4	77.4	77.2	74.3	68.5	64.4
		MCDB	70.5	73.4	80.0	82.5	90.7	92.7	92.3	89.9	90.8	83.6	75.5	70.6
	2%	WB	56.2	62.5	66.0	69.4	74.6	77.1	77.2	76.0	75.4	72.4	64.7	59.6
		MCDB	65.4	72.1	77.0	80.5	87.0	91.1	91.5	90.8	86.6	81.8	71.5	65.5
	5%	WB	52.3	60.0	63.9	68.1	73.1	75.5	76.2	75.1	74.2	70.2	62.8	55.5
		MCDB	63.0	68.3	73.2	78.7	84.6	88.8	90.9	90.5	85.1	78.4	70.4	62.7
	10%	WB	49.3	55.9	61.6	66.3	71.6	74.5	75.2	74.2	73.2	67.9	60.0	51.2
		MCDB	59.8	65.3	70.8	76.1	83.0	88.0	89.8	89.3	84.0	76.6	68.5	60.1
(35) Mean Daily Temperature Range	5% DB	MDBR	23.2	22.7	23.8	23.6	21.9	21.6	22.1	22.5	20.9	22.5	23.0	20.3
		MCDBR	31.8	30.7	29.4	29.5	26.9	24.9	25.5	26.1	24.7	27.1	28.0	28.6
	5% WB	MCWBR	16.9	15.7	13.7	12.9	9.3	6.4	5.6	5.9	6.3	11.1	14.7	16.3
		MCWBR	28.0	28.1	23.7	22.5	21.8	20.9	21.0	22.0	18.9	20.3	22.3	23.7
(40) Clear-Sky Solar Irradiance	taub		0.294	0.304	0.338	0.373	0.412	0.427	0.426	0.419	0.387	0.353	0.317	0.298
		taud	2.520	2.503	2.423	2.351	2.312	2.307	2.348	2.314	2.380	2.461	2.495	2.507
	Ebn at Noon		294	302	298	290	278	273	272	272	278	280	282	285
		Edh at Noon	26	30	35	39	41	42	40	40	36	31	27	25
(44) All-Sky Solar Radiation	RadAvg		953	1140	1517	1840	1974	2185	2187	2002	1665	1296	1005	820
	RadStd		98	167	94	149	205	188	137	116	140	187	126	94

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page